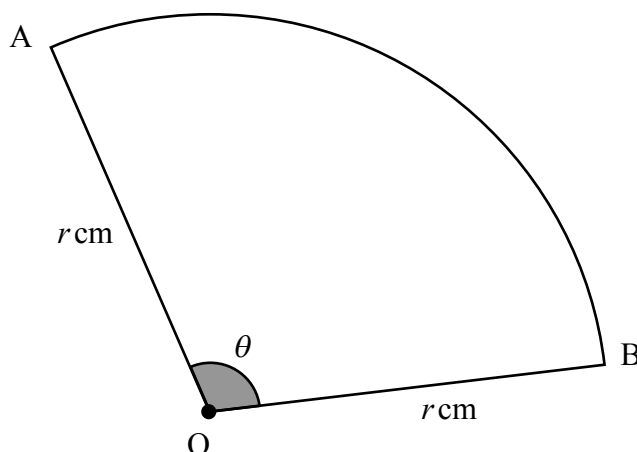


3. [Maximum mark: 8]

Points A and B lie on the circumference of a circle of radius r cm with centre at O.

The sector OAB is shown on the following diagram. The angle $\widehat{AOB}$ is denoted as θ and is measured in radians.

diagram not to scale



The perimeter of the sector is 10 cm and the area of the sector is 6.25 cm^2 .

- (a) Show that $4r^2 - 20r + 25 = 0$. [4]
- (b) Hence, or otherwise, find the value of r and the value of θ . [4]

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